

PHYS 230 - Electricity and Magnetism

Lecture 2

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

1 Electric Charge and Electric Field

- Vectors
- Coulomb's Law

All figures in these slides are from one of the following sources, unless cited otherwise:

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- **Scalars:** Only have **magnitude**, e.g., temperature, length
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- Tensors (not our concern): matrices

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F

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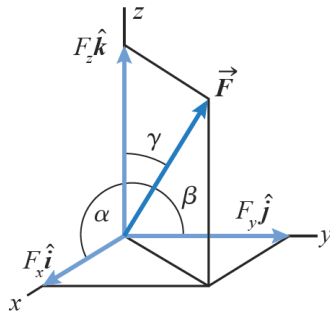


Figure 2-13 Coordinate direction angles, α , β , γ , describing the orientation of vector $\vec{F}$ relative to coordinate axes x , y , and z .

Vectors

Vectors can be thought of arrows with length (magnitude) and direction described by

- Their length and direction, or
- Their components

$$\mathbf{F} = F_x \hat{\mathbf{i}} + F_y \hat{\mathbf{j}} + F_z \hat{\mathbf{k}}$$

or

$$\mathbf{F} = (F_x, F_y, F_z)$$

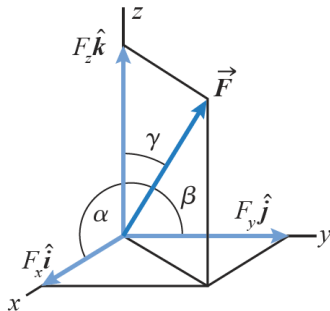


Figure 2-13 Coordinate direction angles, α , β , γ , describing the orientation of vector $\vec{F}$ relative to coordinate axes x , y , and z .

Magnitude of Vectors

Length or magnitude of a vector $F_x\hat{\mathbf{i}} + F_y\hat{\mathbf{j}} + F_z\hat{\mathbf{k}}$ is

$$F = |\mathbf{F}| = \sqrt{F_x^2 + F_y^2 + F_z^2}$$

Unit Vectors

We can construct a corresponding unit vector $\hat{\mathbf{F}}$ out of any vector $\mathbf{F} = F_x\hat{\mathbf{i}} + F_y\hat{\mathbf{j}} + F_z\hat{\mathbf{k}}$ as

$$\begin{aligned}\hat{\mathbf{F}} &= \frac{\mathbf{F}}{F} \\ &= \frac{(F_x, F_y, F_z)}{\sqrt{F_x^2 + F_y^2 + F_z^2}} \\ &= \left(\frac{F_x}{\sqrt{F_x^2 + F_y^2 + F_z^2}}, \frac{F_y}{\sqrt{F_x^2 + F_y^2 + F_z^2}}, \frac{F_z}{\sqrt{F_x^2 + F_y^2 + F_z^2}} \right)\end{aligned}$$

Exercise I <https://epoll.srv.ualberta.ca> – Code: MBD

Prove that the magnitude of any unit vector is 1.

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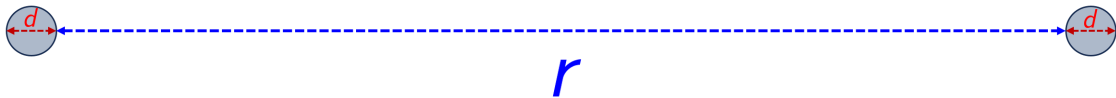
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Point Charge

POINT CHARGE

Charged objects whose dimensions are much smaller in comparison with the distance r between them

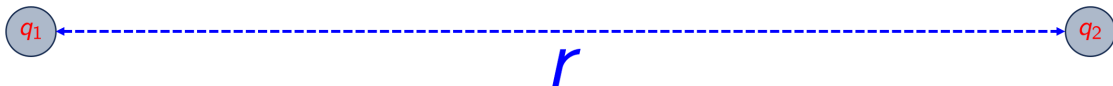


$$r \ll d$$

Coulomb's Law

The magnitude of the electric force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them

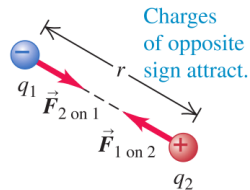
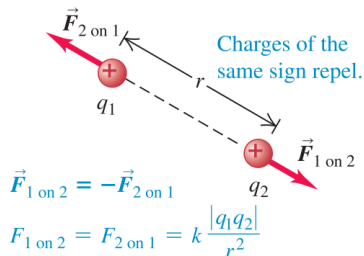
$$F = k \frac{|q_1 q_2|}{r^2}$$



Coulomb's Law

- The directions of the forces the two charges exert on each other are always along the line joining them

$$F_{12} = F_{21} = k \frac{|q_1 q_2|}{r^2}$$

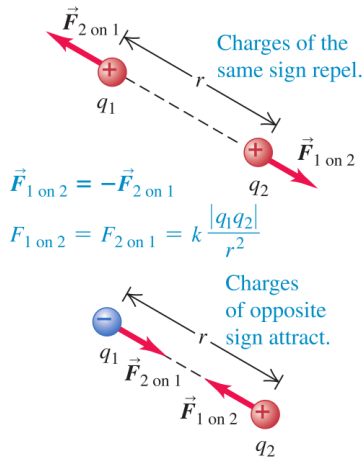


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- The two forces obey Newton's third law: they are always equal in magnitude and opposite in direction, even when the charges are not equal in magnitude.



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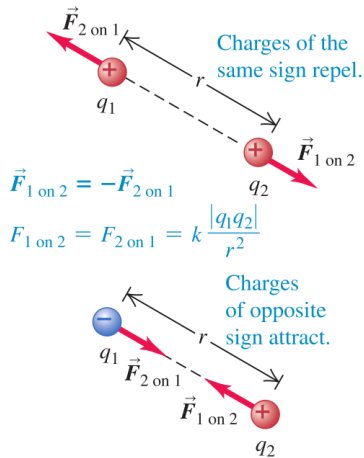
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- k is the is a proportionality constant whose numerical value depends on the system of units used. In SI:

$$k = 9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$$



Coulomb's Law

In SI units we usually write the constant k as

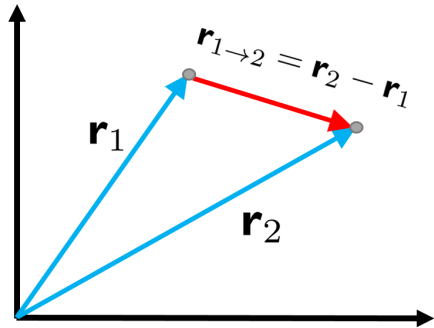
$$k = \frac{1}{4\pi\epsilon_0}$$

where ϵ_0 is called the **vacuum permittivity** or **electric constant**.

Electric Force Vector

The electric force can be written in vector form as

$$\mathbf{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} \hat{\mathbf{r}}_{12}$$

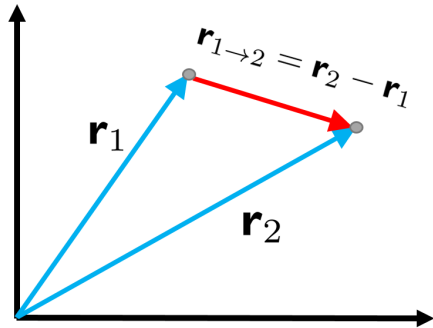


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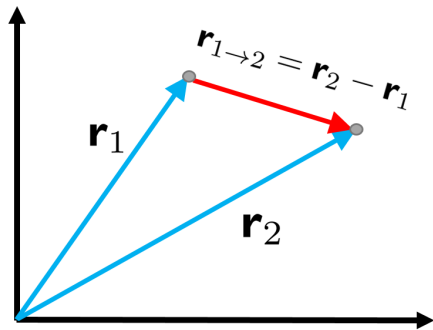
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$$\hat{\mathbf{r}}_{12} = \frac{\mathbf{r}_{12}}{r_{12}} = \frac{\mathbf{r}_2 - \mathbf{r}_1}{r_{12}}$$



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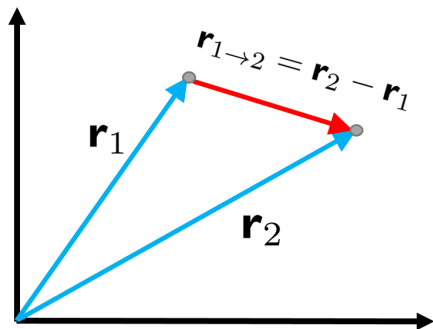
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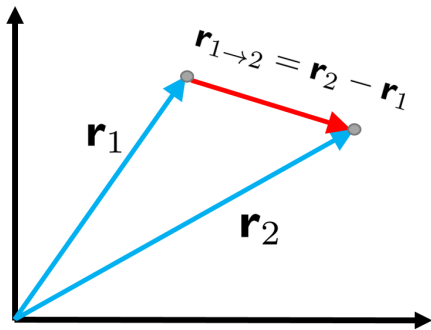
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- $r_{12} = r_{21}$: the distance between the centers of q_1 and q_2
 - Note: $r_{12} = r_{21} = |\mathbf{r}_{12}| = |\mathbf{r}_{21}|$



Electric Force Vector

- The electric force vector can also be explicitly written as

$$\mathbf{F}_{12} = \frac{1}{4\pi\epsilon_0} q_1 q_2 \frac{\mathbf{r}_2 - \mathbf{r}_1}{|\mathbf{r}_2 - \mathbf{r}_1|^3}$$

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- If $q_1 q_2 < 0$ (if the two charges have different signs) $\mathbf{F}_{12}$ is in the direction of $-\hat{\mathbf{r}}_{12}$: attraction

Equivalent Forms of Electric Force Vector

Depending on the problem, one of the three equivalent form of vector electric field is used:

Form 1: If you only need the magnitude (or you can find the direction with other methods)

$$F_{12} = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r_{12}^2}$$

Form 2: Computing the unit vector $\hat{\mathbf{r}}_{12}$ and distance r separately

$$\mathbf{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} \hat{\mathbf{r}}_{12}$$

Form 3: Computing everything together

$$\mathbf{F}_{12} = \frac{1}{4\pi\epsilon_0} q_1 q_2 \frac{\mathbf{r}_2 - \mathbf{r}_1}{|\mathbf{r}_2 - \mathbf{r}_1|^3}$$

Principle of Superposition

The total or net (vector) electric force on any object, say, q_1 , is the vector sum of the individual forces acting on it by other charges $q_2, q_3, \dots$

$$\mathbf{F}_1 = \mathbf{F}_{21} + \mathbf{F}_{31} + \mathbf{F}_{41} + \dots$$

Strategy To Find Electric Force Vector/Magnitude

1. Choose a coordinate system wisely
2. Depending on the problem, choose the formula (forms 1, 2, 3)
3. Divide computations into simpler steps, e.g., first find $\mathbf{r}_2 - \mathbf{r}_1$, then $|\mathbf{r}_2 - \mathbf{r}_1|$, etc.
4. Find $\mathbf{F}$ for each point charge; if necessary use the superposition principle to find the net force

Exercise 2

<https://epoll.srv.ualberta.ca> – Code: MBD

Two point charges, $q_1 = 25 \text{ nC}$ and $q_2 = -75 \text{ nC}$ are separated by a distance $r = 3 \text{ cm}$. Find the magnitude of the electric force

(a) that q_1 exerts on q_2 , and

(b) that q_2 exerts on q_1

Exercise 3

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Two point charges are located on the x -axis of a coordinate system: $q_1 = 1.0 \text{ nC}$ is at $x = +2.0 \text{ cm}$, and $q_2 = -3.0 \text{ nC}$ is at $x = +4.0 \text{ cm}$. What is the total vector electric force exerted by q_1 and q_2 on a charge $q_3 = 5.0 \text{ nC}$ located at $x = 0$?

Exercise 3

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